Background Survey

Project: AI-Assisted Travel Planning Web Application

1. Purpose of the Survey

This background survey reviews three existing travel platforms - Trip.com, Expedia and Booking.com - to identify common interface patterns, useful features and design ideas relevant to the proposed AI-assisted travel planning web application.

The purpose is not to reproduce these commercial services. Instead, the survey is used to understand how established travel products collect user requirements, organise travel information and reduce the amount of effort required from users during trip planning.

2. Platforms Reviewed

- Trip.com - a broad travel platform covering hotels, flights, trains, cars, attractions and travel packages.
- Expedia - a search-led travel platform covering stays, flights, cars, packages and activities.
- Booking.com - an accommodation-led travel platform that also provides flights, car rental, attractions and other travel services.

Screenshots of the three home pages were captured and retained as evidence for the project background survey.

3. Key Interface Observations

Trip.com

- Provides a wide set of travel categories in a clearly visible navigation area.
- Uses a prominent central search/planning area.
- Collects a small number of high-value inputs such as destination, dates and traveller information before presenting more detailed results.
- Uses cards, icons and category grouping to make a large amount of travel information easier to scan.

Expedia

- Uses a simple search-first layout with destination, dates and travellers as the main inputs.
- Allows users to move between travel categories without changing the overall interaction pattern.
- Keeps the initial decision process relatively simple, with detailed choices appearing after the first search.
- Uses a clear primary search action and strong visual hierarchy.

Booking.com

- Keeps accommodation search as the main focus while also providing access to flights, car rental, attractions and other travel services.
- Uses destination, dates and occupancy as the core initial inputs.
- Places the main search controls together in one compact area.
- Uses clear navigation and familiar booking patterns that reduce the amount of explanation required for the user.

4. Cross-Platform Findings

Although the three platforms have different commercial priorities, several common UX patterns were identified:

- Ask for only a few important pieces of information at the beginning.
- Use structured inputs rather than requiring users to describe the whole trip in free text.
- Make destination and travel dates highly visible.
- Use a strong primary call-to-action to continue the planning process.
- Present information using cards, categories and clear visual grouping.
- Reveal more detailed choices progressively instead of placing everything on the first screen.
- Support mobile-friendly interaction and simple scanning of travel information.

5. Implications for the Proposed System

The survey supports a structured-input approach for the proposed system. The first screen should collect only the information that is most useful for itinerary generation:

- destination;
- travel dates;
- budget;
- number of travellers;
- personal interests;
- optional additional preferences.

The system should then generate a personalised itinerary and present it in a clear day-by-day format instead of relying on a long chatbot response.

Unlike the reviewed commercial platforms, the project does not aim to provide a complete booking service. Its focus is AI-assisted itinerary planning, structured presentation, budget awareness and clearer communication of uncertain information.

6. Design Direction Derived from the Survey

- Use a clean and prominent trip-planning form inspired by established travel search patterns.
- Keep the first interaction simple and avoid excessive options.
- Use cards and short sections for the generated itinerary.
- Show budget information separately from activity descriptions.
- Use responsive layouts so that the planner remains usable on mobile devices.
- Avoid commercial features such as advertisements, loyalty schemes, payment and booking checkout in the initial project scope.

7. Preliminary Conclusion

The survey suggests that established travel products reduce user effort by combining simple structured input with clear information hierarchy. This supports the proposed design direction for the project: structured travel preferences should be collected first, followed by an AI-generated itinerary presented in a focused and easy-to-scan interface.